

B&W Productions — Wages System Handover & Restore Point

Rate rules · Review colours · Missed shifts · Payroll Excel for the auditor

15 September 2026

1. Purpose of this document

This is the **restore point** for the wages system as approved by Bernie Burness on 15 September 2026. If anything ever breaks or is changed by mistake, this document (plus the git tag and backup listed in §12) describes **exactly** how the system must look and behave: the pay rules, the colours and labels on the dashboard, how overlaps and reviews appear, how missed shifts are handled, and how the Payroll Excel for the auditor is laid out.

Item	Value
Live site	https://bwprodssystem.co.za (staff: /wages, office: /admin/wages)
Code version live	v2026-09-15-10 (check at /wages-version)
Cloudflare project	bw-productions (branch main = production, realdate-test = preview)
Database	Cloudflare D1 bw-productions-db
Git restore tag	restore-2026-09-15-rate-rules-payroll-excel
Previous restore point	restore-2026-09-14-combined-sheet (commit 807323c)

2. What changed on 15 September 2026 (in order)

1. **Final Shift Check** shows the real date of a missed shift (not Saturday); **one tick box** instead of three.
 2. **Select all → Submit selected** now really final-submits every ticked draft; skips “Submitted — locked” and out-of-window cards.
 3.  **Edit** on every row of the combined sheet (paid rows → engine correction form; drafts → /wages-admin/edit-draft/:id).
 4. **Calendar window**: a worker can only capture from **last payroll’s Saturday to this payroll’s Friday**. Older/newer dates are refused on the server and greyed out on the calendar.
 5. **New B&W rate rules** (§3) applied automatically at final submission, **with the midnight split** (§3.4).
 6. **Existing week (12–18 Sep) corrected** to the new rules — 14 rows changed, total R22 791,25 → R20 751,25 (before-state saved in docs/reprice-backup-2026-09-15-week-2026-09-12-before.json).
 7. **Rate-choice review**: “warehouse” in the wording but work type not Warehouse → Bernie chooses Warehouse or Event/Venue before the rate is applied (§5.4).
 8. John’s two overlapping Sunday shifts put into red Review (not deleted); Erick’s Monday “warehouse loading” reverted to R750 pending Bernie’s choice.
 9. **Payroll Excel for the auditor** — new 6-tab workbook built from the database (§8). Button “Download Payroll Excel (new)” on /admin/wages.
 10. Excel layout tidy-up (merged title bands, one-line rows, aligned loans grid).
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3. B&W pay rules (owner, 15 Sep 2026) — OWNER_RATE_RULES_VERSION = 4

All prices are worked out **per shift, on the actual day the hours were worked**, after the worker final-submits. The engine's own amount is overwritten by the proxy and the row is stamped `calculation_version = 4` with a note beginning `Rate rules 2026-09-15: ...`.

3.1 General staff

Day	Warehouse (work type <i>Warehouse Team</i>)	Event / Venue (everything else)
Mon–Fri	R81,25 per hour	R750 fixed for any time inside 07:00–16:00 + R90/h for every hour before 07:00 or after 16:00
Saturday	R95 per hour	R95 per hour
Sunday	R120 per hour	R120 per hour

3.2 Music Bus (work type *Music Bus*)

Day	Rule
Mon–Sat	R750 fixed for time inside 07:00–16:00 + R120/h outside
Sunday	R120 per hour

3.3 Petrus (staff #4, Tsotlego Petrus Malakoane)

Day	Rule
Mon–Fri	R650 fixed for time inside 07:00–16:00 + R81,25/h outside
Sat & Sun	R95 per hour

3.4 Midnight split (owner instruction)

Use the actual day the hours were worked. If a shift goes past midnight, the hours after midnight use the new day's rate. It stays **one shift on one row**; only the pay calculation splits.

Example — Friday 22:00–02:00 Event: 22:00–00:00 = Friday (2 h × R90 = R180) + 00:00–02:00 = Saturday (2 h × R95 = R190) = **R370**. Note on the row: "... (until midnight) + Saturday: 2 h × R95 (after midnight, Sat)".

3.5 Gardeners (staff #1 Givemore, #2 Takavaudza)

House / House/Garden block keeps the engine's **gardener rate (R62,50/h)** — not recalculated. Their Team block (Warehouse Team / Team Assistance) is priced as general staff.

3.6 Fixed weekly staff (Sharleen) — not recalculated.

3.7 Never again

- No R375 half day. * No old Sunday ×1.5 / ×1.2 factors. * No Music Bus R130 or R190.

3.8 Worked examples (all pass in the unit test)

Shift	Amount
Warehouse Mon 08:00–16:00	R650

Shift	Amount
Event Mon 05:00–16:00	R930 (750 + 2 × 90)
Event Sun 07:00–12:00	R600
Music Bus Sat 08:00–22:00	R1 470 (750 + 6 × 120)
Petrus Fri 20:00–01:00	R420 (4 × 81,25 + 1 × 95 Sat)
Event Sat 22:00–02:00	R430 (2 × 95 + 2 × 120 Sun)
Event Sun 18:00–00:00 (ends exactly midnight)	R720 — no split

4. Payroll week and calendar window

- Payroll week runs **Saturday → Friday**.
- Workers may capture dates from **last payroll's Saturday** up to **this payroll's Friday** only. Anything older is a red card "**OLDER THAN ONE WEEK — cannot be final-submitted. Delete it or speak to the office.**"; anything later is "**NEXT PAYROLL — capture it from Saturday.**" Both are refused by the server as well.
- A shift from the previous payroll week captured now is a **missed shift**: it keeps its **real date** and is paid in the **current** payroll (`payroll_week_start` = this Saturday, `missed_previous_week` = 1).

5. The office dashboard /admin/wages — combined sheet

5.1 Person header line (left → right)

Element	Colour	Meaning
NAME Isaac Mbele	white bold	worker
76.00 hours · R6 570,00	yellow/gold #e2b93b	Claimed — final-submitted and paid by the engine, before review decisions
Admin approved: 58.00 hours · R5 013,75	green #86efac	Approved — claimed less what Bernie's resolved reviews took off
Admin approved: 3 reviews still open	amber #fbbf24	approved figure withheld until every open review has a decision
Open Isaac's worker app ↗	gold button	opens that worker's app in a new tab (office)

5.2 Pills on the header line

Pill text	Background	Text	Meaning
☞ all days match pay rule / ☞ rule ok	dark green #1f5f3a	white	pay-rule check agrees with what was paid
☞ RULE +R500,00	dark red #7f1d1d	white	paid more than the rule says (click to open detail)
☞ RULE -R75,00	dark blue #1e3a8a	white	paid less than the rule says
▮ 9 open reviews ▾	orange #b45309	white	this person has open reviews (click to open list)
▮ REVIEW	dark red #7f1d1d	white	red (action-needed) review on this person

Pill text	Background	Text	Meaning
2 missed shifts · 26.00 h · R2 220,00	pink #fdecec	dark red #7f1d1d	missed shifts from the previous payroll paid in this one
fixed weekly – rule check not applied	grey #374151	white	Sharleen

5.3 Row pills (per shift)

Pill	Colours	Meaning
MISSED	pink / dark red	real date is before this payroll's Saturday
⚙ rule ok	dark green / white	this day's paid amount matches the rule
📄 review	orange / white	review open on this shift
NOT FINAL-SUBMITTED	grey #374151 / white	draft — counts as R0 until final-submitted
next day	dark blue / white	overnight shift ends the next day
corrected	dark green / white	manager correction recorded

5.4 Review column — what the system shows

Open review (orange) REVIEW – open #9770 > Overlap check — 2026-09-13 > Manual overlap / possible already-claimed review: the same employee has another overlapping entry on 2026-09-13 for 07:00–12:00. > Recommended payable: 2.00 h (05:00–07:00) · already paid 7.00 h that day ← *only when the system can work it out* > Reason to record: “07:00–16:00 on 10 Sep already paid in payroll 2026-09-05; only the early hours 05:00–07:00 are payable.” > [Approve hours ____] [Reason _____] [Record decision] [Pay as claimed (9.00 h)] [No issue – pay as claimed]

Action-needed review (dark red) REVIEW – ACTION NEEDED #9758 * **Overlapping paid shifts** — two paid rows for the same time (e.g. John Sun 13 Sep 10:00–12:00 ×2). Text: “TWO OVERLAPPING PAID SHIFTS ... Both are currently paid; neither has been deleted or changed. Bernie to decide: pay both, pay one (approve 0 h on the other), or adjust hours.” * **Rate choice** — wording says “warehouse” but the work type is not Warehouse. Text: “RATE CHOICE NEEDED: ... Bernie must choose Warehouse (R81,25/h weekday) or Event/Venue (R750 fixed 07–16 + R90/h outside) before the rate is decided. Until then the shift stays at the engine amount.” Buttons: [Warehouse (R81,25/h)] [Event/Venue (R750 fixed 07–16 + R90/h outside)] → re-prices only that shift, records Bernie's name.

Decided review (green text) REVIEW – RESOLVED #9665 > **Decided: 1.00 h** by Bernie Burness — Already built in North payroll. The Excel sheet shows this, which I've saved on file on the PC.

The review decision is stored on wage_payroll_reviews (status RESOLVED, approved_payable_hours, decision_reason, reviewed_by_name, reviewed_at). **It does not change the engine's paid row**; the green “Admin approved” figure and the Payroll Excel use the approved hours.

5.5 Missed-shift detail column

MISSED SHIFT – actual date Wed 9 Sep 2026 – Set up HQ (pink pill)
 Worked Wed 9 Sep 2026 (previous payroll), not captured that week;
 final-submitted and paid in payroll 2026-09-12 to 2026-09-18.
 CLEAR worked 07:00–16:00 that day (paid); these hours are outside those times (green pill)
 – or –
 OVERLAP already paid 05:00–16:00 (Grosvenor, Sandton – setup, payroll 2026-09-05) (pink pill)
 🛠 Edit shift

5.6 Grand total tiles (bottom of the sheet)

GRAND TOTAL (incl. missed shifts) – 47 paid shifts · R28 965,01 (yellow, claimed) and **Agreed after reviews** in green (amber while any review is open).

6. Worker app /wages — what staff see

- Calendar limited to the window in \$4; message under the picker explains the window.
- Missed-shift card: pink pill **MISSED SHIFT – actual date Thu 10 Sep 2026 – Strike HQ Evening**.
- Final Shift Check: real date shown; **one tick box** (“The times, end time and information are correct”) replaces the three.
- **Select all → Submit selected** final-submits every ticked draft; locked and out-of-window cards are skipped and reported.
- After final submission the shift card shows **Submitted — locked**.

7. Missed shifts — the rule

1. Worker captures the shift with its **real date** (previous week).
2. It is paid in the **current** payroll; the dashboard shows it under the worker with the MISSED pill and the CLEAR / OVERLAP cross-check against what was already paid that day in the earlier payroll.
3. If OVERLAP, a review opens; **Bernie decides the approved hours** (the green figure).
4. The **auditor only ever sees the approved hours** (Excel block 6). The claimed hours and the reasoning stay on the dashboard and on the Missed Shifts tab notes, for answering staff questions.

8. Payroll Excel for the auditor — `/wages-admin/payroll.xlsx?from=YYYY-MM-DD`

Button “**Download Payroll Excel (new)**” on /admin/wages, next to the old engine export (relabelled “*Export Excel (old rules – reference only)*” — it uses the old formulas and excludes missed shifts; kept for comparison only, to be removed later).

Six tabs, in this order:

Tab 1 · Wage Detail Linked (master — edit only the yellow cells)

Shift ID · Employee · Date · Day · Missed shift? · Venue / Outlet · Area · Description · Work type · Start · End · Claimed hours · Approved hours (system) · Override hours (edit) · Effective hours · Amount (system) · Override amount (edit) · Effective amount · Rate used · Breakdown · Override reason (edit) · Review # · System note · Review decision One row per paid shift **including missed shifts** (real date). Approved hours = the green review decision, or claimed where no review. Amount = new rate rules on the approved hours. Grand total row at the bottom.

Tab 2 · Auditor Trail Linked

[illegible]

Fixed Deductions Due | Fixed Deducted (edit) | Fixed Outstanding After Payroll |
 Additional Loan Due | Additional Loan Deducted (edit) | Additional Loan Outstanding After Payroll |
 Total Deducted This Period | NET WAGE

- **No Bonus column, no “Gross Including Bonus”.**
- Sat-Fri cells are SUMIFS over Tab 1 (effective hours / effective amount, in-week rows only).
- **6 MISSED SHIFTS** = each worker’s missed total pulled from Tab 6 (approved hours).
- Total Hours = Sat + Sun + Mon + Tue + Wed + Thu + Fri + Missed.
- Yellow “(edit)” cells for what was actually deducted; Net = Wages – Total Deducted.
- GRAND TOTAL row.

Tab 3 • Auditor Summary

Name • Days Worked • Hours (incl. missed) • Missed Shift Hours • Wages • Fixed Deductions Due / Deducted / Outstanding • Additional Loan Due / Deducted / Outstanding • Total Deducted •
 NET WAGE — every cell linked to Tab 2. Grand total.

Tab 4 • Flagged – Reviewed

Status • Review # • Employee • Date • Shift • Flag • System note • System recommended hours • Claimed hours • Approved hours • Decision • Decided by • Decided at • Reason recorded — OPEN rows first (pink status cell).

Tab 5 • Loans & Deductions

Three blocks on one aligned grid: **Fixed weekly deductions** (Erick Car R1 000 • Isaac Eggs R260 • Petrus Rent R450); **Additional loans** (loan #, dates, reason, original, instalment, repaid, **outstanding balance**, due this week, status, **Check** column — flags e.g. “Marked PAID but ledger shows only R0,00 repaid of R3 000,00 — please check”); **Loan repayment history** (all payrolls).

Tab 6 • Missed Shifts

Worker • Missed shift • Approved hours • Amount • Notes

Isaac Mbele	Wed 9 Sep 2026 07:00–17:00	1.00	R90,00	
Notes: Worked Wed 9 Sep 2026 (previous payroll), not captured that week; final-submitted and paid in payroll 2026-09-12 to 2026-09-18.				
OVERLAP already paid 05:00–16:00 (Grosvenor, Sandton – setup, payroll 2026-09-05)				
Isaac Mbele	Thu 10 Sep 2026 07:00–23:00	7.00	R630,00	
Notes: Worked Thu 10 Sep 2026 ... OVERLAP already paid 07:00–16:00 (Grosvenor, Sandton – strike, payroll 2026-09-05)				
Isaac Mbele – missed total		8.00	R720,00	← feeds block 6 on Tab 2
...				
GRAND TOTAL MISSED SHIFTS		69.00	R6 330,00	

Approved hours and Amount are linked to Tab 1; a change there flows to Tab 6 → Tab 2 → Tab 3.

Formatting standard

Title row (bold 13) and one grey italic note row, both merged across the sheet; grey header row (bold, centred, wrapped); one-line data rows; yellow #FFF2A8 = editable; light-blue #EAF3FF = sub-totals; pink #FDE2E2 = needs attention; money as R#, ##0.00; header rows frozen. Workbook recalculates on open.

Reference figures for week 2026-09-12 → 2026-09-18 (as at 15 Sep 12:40)

47 paid shifts (18 missed) · claimed 315,5 h · **approved 260,75 h** · missed approved 69 h · wages **R27 198,76** · deductions R3 210 · **net R23 988,76** · 0 open reviews on paid rows.

9. Corrections applied to week 12–18 Sep (audit trail)

Worker	Shift	Old	New	Rule
John	Sat 12 · 09:00–11:00	R750	R190	Sat 2 h × R95
John	Sat 12 · Music Bus 11:00–20:00	R1 170	R1 230	R750 + 4 h × R120
John	Sat 12 · 20:00–22:00	R180	R190	Sat 2 h × R95
John	Sun 13 · 10:00–12:00 (×2)	R270 each	R240 each	Sun 2 h × R120 — in red review
John	Sun 13 · Music Bus 12:00–19:00	R1 330	R840	Sun 7 h × R120
Erence	Sun 13 · 07:00–13:00	R810	R720	Sun 6 h × R120
Erick	Sun 13 · 07:00–12:00	R675	R600	Sun 5 h × R120
Erick	Mon 14 · “warehouse loading”	R750	R750 (reverted)	Bernie chose Warehouse → R731,25 via review #9758
Solomon	Sat 12 · 07:00–12:00	R750	R475	Sat 5 h × R95
Solomon	Sun 13 · 07:00–12:00	R675	R600	Sun 5 h × R120
Patrick	Sat 12 · 06:45–11:45	R772,50	R475	Sat 5 h × R95
Patrick	Sun 13 · 07:00–12:15	R708,75	R630	Sun 5,25 h × R120
Joshua	Sun 13 · 07:00–13:00	R810	R720	Sun 6 h × R120

Each corrected row carries the note “Rate rules 2026-09-15 (admin correction of existing row, was R...): ...”. Before-state JSON: `docs/reprice-backup-2026-09-15-week-2026-09-12-before.json`.

10. Database objects the proxy relies on (no schema changes were made to engine tables)

Table	Used for
<code>wage_shifts</code>	paid rows; proxy updates <code>total_amount</code> , <code>gross_wage</code> , <code>hourly_rate_snapshot</code> , <code>calculation_version</code> = 4, appends to <code>payroll_note</code>
<code>wage_shift_drafts</code>	drafts; <code>payroll_week_start</code> , <code>missed_previous_week</code>
<code>wage_payroll_reviews</code>	reviews; proxy inserts <code>issue_key</code> = <code>rate_choice_warehouse_or_event\ shift:ID</code> (severity red) and records decisions
<code>wage_draft_real_dates</code> (proxy)	real date parked while the engine is told “Saturday”
<code>wage_shifts_removed_audit</code> (proxy)	copies of rows removed by the office
<code>wage_debug_capture</code> (proxy)	log of every re-price, decision and rate choice

Table	Used for
wage_recurring_deductions, wage_additional_loans, wage_additional_loan_repayments	Excel tab 5
Proxy routes added: GET /wages-admin/payroll.xlsx, POST /wages-admin/rate-choice, POST /wages-admin/review-decision, GET /wages-admin/edit-draft/:id, GET /wages-version, GET /wages-health.	
Source files: src/index.tsx (proxy), src/payroll-excel.ts (workbook), src/xlsx-lite.ts (Excel writer).	

11. Open items agreed to look at later

1. Void the 4 false “self-compare” reviews (a draft compared with its own paid row): Isaac 18 Sep #9762, Petrus #9759/#9760/#9761 — **awaiting Bernie’s yes.**
2. Drafts’ open reviews should not block the green “Admin approved” figure — **awaiting yes.**
3. Dashboard totals still show *claimed*; Excel shows *approved*. Bringing the dashboard in line is a separate instruction.
4. Daniel’s R3 000 loan is marked paid with R0 in the ledger — flagged on the Loans tab.
5. Remove the CSV button and the old “reference only” export once the new Excel has been used for a full payroll.
6. Erick’s open reviews #9690 / #9691.

12. How to restore this exact state

```
git checkout restore-2026-09-15-rate-rules-payroll-excel
npm run build
npx wrangler pages deploy dist --project-name bw-productions --branch main --commit-dirty=true
curl https://bwprodsystem.co.za/wages-version # must return v2026-09-15-10
```

Backup archive of the whole project (code + docs + this file) is listed in the README under “Backups”. Data corrections can be reversed from the before-state JSON in docs/.

Prepared for B&W Productions — Bernie Burness — 15 September 2026.